

MCQ-6

Topic: Probability

Student Name: _____ Date: _____ Score: _____

Instructions: Select the correct option by filling or marking the checkbox ☐ next to it.

1. If an event is absolutely certain to happen, its probability is:

- ☐ 0.25
- ☐ 0.5
- ☐ 0.75
- ☐ 1

2. A fair six-sided die is rolled once. What is the probability of rolling an even number?

- ☐ 1/6
- ☐ 1/3
- ☐ 1/2
- ☐ 2/3

3. If $E(X) = 5$ and $E(Y) = 3$, and X and Y are independent random variables, what is $E(2X - 3Y)$?

- ☐ 1
- ☐ 4
- ☐ 7
- ☐ 19

4. Two events A and B are mutually exclusive. Which of the following is always true?

- ☐ $P(A \cap B) = P(A)P(B)$
- ☐ $P(A \cup B) = P(A) + P(B) - P(A)P(B)$
- ☐ $P(A \cup B) = P(A) + P(B)$
- ☐ $P(A | B) = P(A)$

5. If A and B are mutually exclusive and both have positive probability, then:

- ☐ They are independent
- ☐ They are dependent
- ☐ $P(A | B) = P(A)$
- ☐ $P(A \cup B) = 1$

6. Given $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$. Find $P(A | B) - P(B | A)$.

- ☐ 0
- ☐ 0.1
- ☐ 0.2
- ☐ 0.3

7. If $P(A | B) = P(A)$, which must be true?

- ☐ A and B are mutually exclusive
 - ☐ A and B are independent
 - ☐ $P(B | A) = 1$
 - ☐ $P(A \cup B) = P(A) + P(B)$
8. A disease affects 1% of the population. A test is 95% accurate for both diseased and non-diseased people (sensitivity = specificity = 0.95). If a person tests positive, the approximate probability that he has the disease is:
- ☐ 0.95
 - ☐ 0.16
 - ☐ 0.50
 - ☐ 0.01
9. Box 1 contains 2 red and 3 blue balls. Box 2 contains 3 red and 2 blue balls. A box is chosen at random and a ball drawn is red. Probability it came from Box 1 is:
- ☐ $2/5$
 - ☐ $3/5$
 - ☐ $2/5 \times 1/2$
 - ☐ $4/5$
10. Which of the following can be a valid PMF of a discrete random variable X taking values 1, 2, 3?
- ☐ $p(1) = 0.4, p(2) = 0.5, p(3) = 0.2$
 - ☐ $p(1) = 0.3, p(2) = 0.4, p(3) = 0.3$
 - ☐ $p(1) = -0.1, p(2) = 0.6, p(3) = 0.5$
 - ☐ $p(1) = 0.5, p(2) = 0.5, p(3) = 0.5$
11. If the mean of a binomial distribution is 4 and variance is 3, then the value of p is:
- ☐ $1/4$
 - ☐ $3/4$
 - ☐ $1/2$
 - ☐ $1/3$
12. If $X \sim \text{Poisson}(\lambda)$ and $P(X = 0) = P(X = 1)$, then λ equals:
- ☐ 0
 - ☐ 1
 - ☐ e
 - ☐ 2
13. Which of the following cannot be a PDF?
- ☐ $f(x) = 2x, \quad 0 < x < 1$
 - ☐ $f(x) = 3x^2, \quad 0 < x < 1$
 - ☐ $f(x) = \frac{1}{2}x, \quad 0 < x < 2$
 - ☐ $f(x) = x, \quad 0 < x < 1$
14. If the PDF of a continuous random variable is $f(x) = 2x$ for $0 < x < 1$, then $P(0.3 < X < 0.7)$ equals:

- ☐ 0.4
☐ 0.41
☐ 0.42
☐ 0.43
15. For a normal distribution, the probability that a value lies within one standard deviation of the mean is approximately:
☐ 0.68
☐ 0.95
☐ 0.98
☐ 0.99
16. $X \sim N(\mu = 50, \sigma = 10)$. $P(X > 65)$ is equal to $P(Z > ?)$ where Z is standard normal.
☐ 1.5
☐ 1.0
☐ 2.0
☐ 0.5
17. If A and B are independent events with $P(A) = 0.3$, $P(B) = 0.4$, then $P(A \cup B)$ is:
☐ 0.12
☐ 0.58
☐ 0.7
☐ 0.18
18. A box has 4 red marbles and 6 blue marbles. If you pick two marbles without replacement, what is the probability that both are red?
☐ $2/15$
☐ $4/25$
☐ $2/5$
☐ $1/3$
19. Three students are selected from a group of 5 boys and 4 girls. Find the probability that all selected students are girls.
☐ $1/22$
☐ $1/21$
☐ $1/20$
☐ $1/19$
20. A die shows an even number. What is the probability that it is greater than 4?
☐ $1/2$
☐ $1/3$
☐ $1/4$
☐ $1/5$